

A top-down view of various cosmetic products including brushes, powders, and creams arranged on a solid blue background. The products are scattered across the left and bottom portions of the frame, leaving the right side mostly empty for text.

Next Frontier for Glossier: Balancing Authenticity with AI-Driven Scale

Consulting Team 5 | May 2025

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Agenda



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Driving Innovation and Growth: Strategic Need for AI Integration at Glossier

Glossier should leverage AI technology to scale its business while preserving the personal, authentic brand experience

Personalization at Scale

AI analyzes user behavior to deliver personalized recommendations at scale while maintaining Glossier's intimate brand feel.

Faster Human-centered Content Creation

AI speeds up content creation and community responses, keeping messaging on-brand and emotionally engaging.



Amplify Real Voices

AI helps surface authentic community stories and content, strengthening brand trust and connection.

Proven Business Impact with AI Adoption

AI has driven up to 400% growth for other brands, and Glossier's 200% gain shows strong potential for continued expansion.

Key Strategic Priorities to Guide Glossier's AI Journey



Implement Strategic AI Initiative

Proceed with the three-phase implementation plan, beginning with the Hybrid Community Management System.



Maintain Human-Centered Approach

Ensure all AI implementations enhance rather than replace authentic human connections.



Establish Governance Structure

Create a formal AI ethics committee and governance framework before implementation begins



Implement Transparent Communication

Develop a customer communication strategy that clearly explains how AI is enhancing the Glossier experience.



Monitor Competitive Landscape

Establish ongoing competitive analysis to track AI implementations across the beauty industry



Prioritize Technical Excellence

Invest in robust data infrastructure and testing to ensure high-quality AI performance



Develop Staff Capabilities

Create comprehensive training programs to ensure staff can effectively collaborate with AI systems.

The Rising Curve of Global Cosmetics

Market
The global cosmetics industry is projected to generate a revenue of USD 445,977.7 million by 2030, growing at a compound annual growth rate (CAGR) of 6% between 2024 and 2030

Revenue, 2023 (US\$M)

\$295,950.2

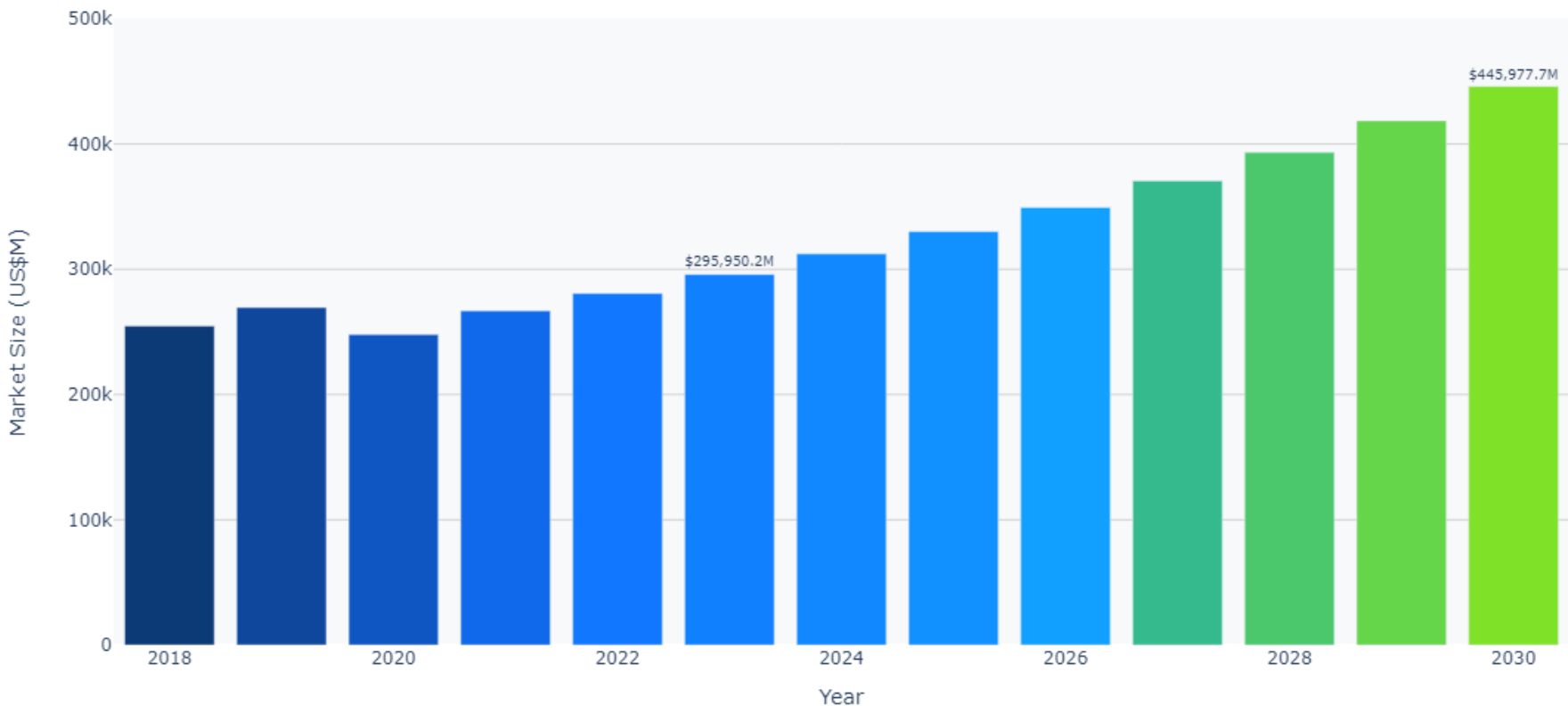
Forecast, 2030 (US\$M)

\$445,977.7

CAGR, 2024 - 2030 (US\$M)

6.0%

Global cosmetics market, 2018-2030 (US\$M)

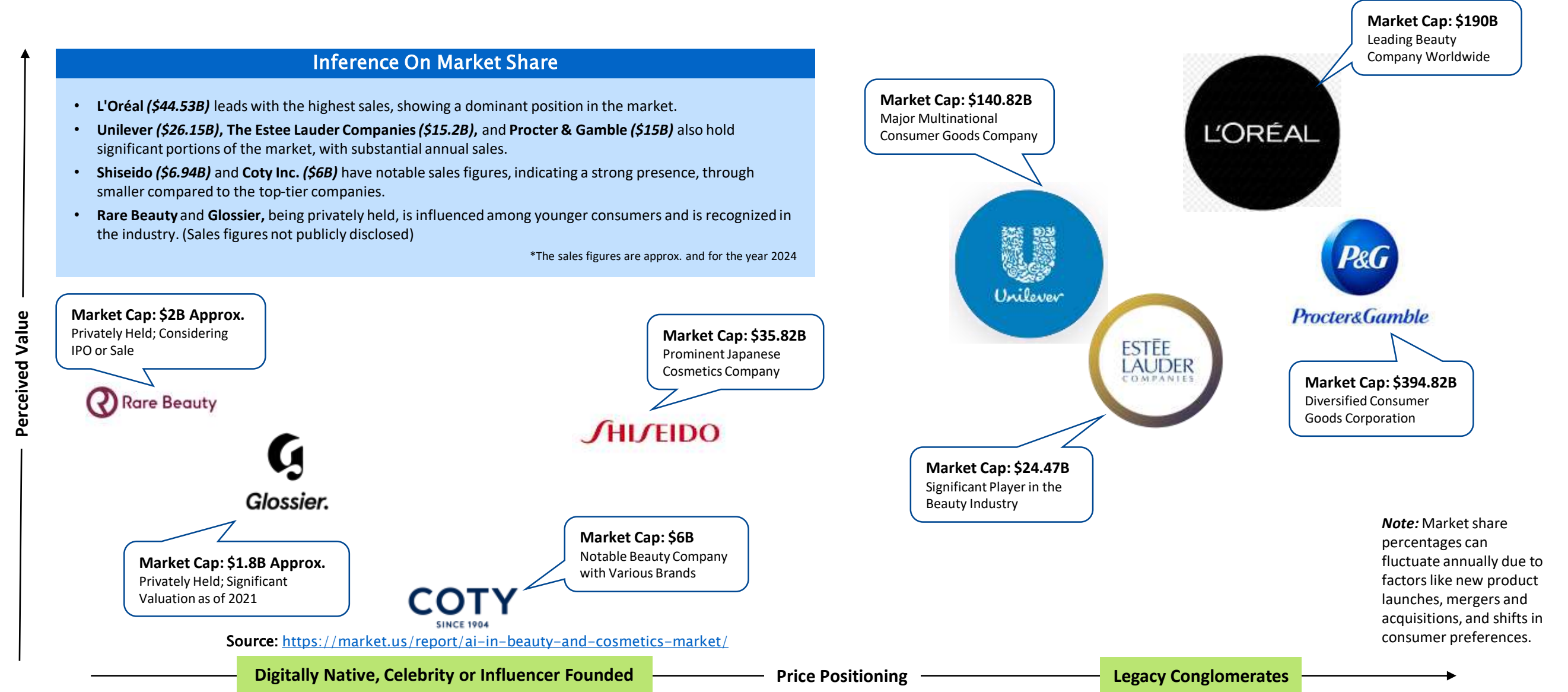


Global Cosmetics Market Insights

- The global cosmetics market earned USD 295,950.2 million in revenue in 2023 and is projected to reach USD 445,977.7 million by 2030.
- From 2024 to 2030, the market is anticipated to grow at a compound annual growth rate (CAGR) of 6%.
- Among all segments, skin care contributed the most in 2023, with a revenue of USD 128,029.1 million.
- Hair Care is the fastest-growing product segment and is considered the most profitable during the forecast period.
- In 2023, the Asia Pacific region led the market in terms of revenue generation.
- On a country level, South Africa is predicted to experience the highest CAGR from 2024 to 2030.

Source: <https://www.grandviewresearch.com/horizon/outlook/cosmetics-market/north-america>

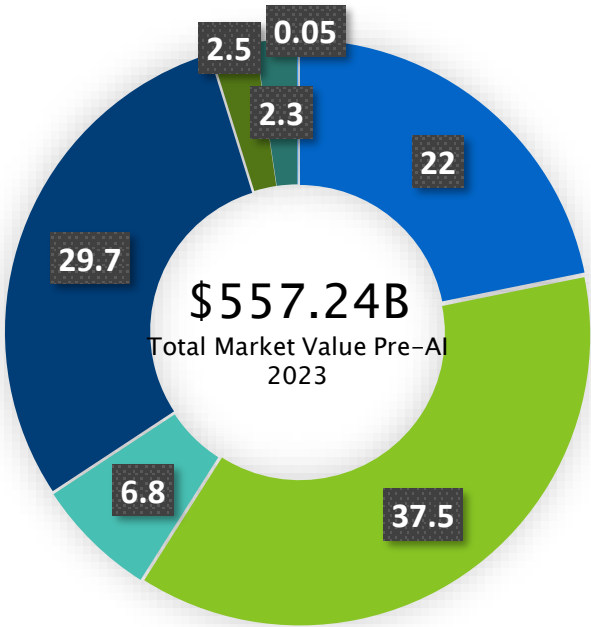
The beauty industry is notably fragmented and competitive, with a handful of major corporations holding significant market share while agile direct-to-consumer (DTC) brands challenge conventional standards. Within this space, Glossier carves out a niche as a minimalist, community-focused brand



The Impact of AI & Technology in Cosmetics Market Growth

The AI in beauty and cosmetics market has experienced rapid expansion in recent years, growing from \$3.72 billion in 2024 to an expected \$4.4 billion in 2025, representing a compound annual growth rate (CAGR) of 18.3%. This growth trajectory is projected to continue, with the market anticipated to reach \$9.44 billion by 2029 at a CAGR of 21.0%

Pre-AI Market Share (Approx. %)



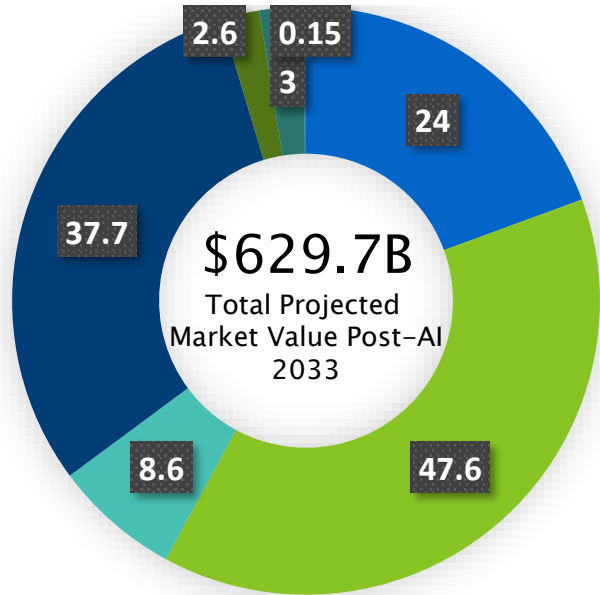
Primary Drivers of Market Expansion

- Personalized Beauty Recommendations
- Advanced Skin Analysis Capabilities
- Virtual Make-Up Applications
- AI-Powered Product Formulation
- Data-Driven Personalization

Major Trends Shaping the Industry

- Color Matching Technologies
- Virtual Try-On Experiences
- AI-Driven Beauty Recommendations
- Augmented Reality Beauty Content Creation

Post-AI Market Share and Projected Value (Approx. %)



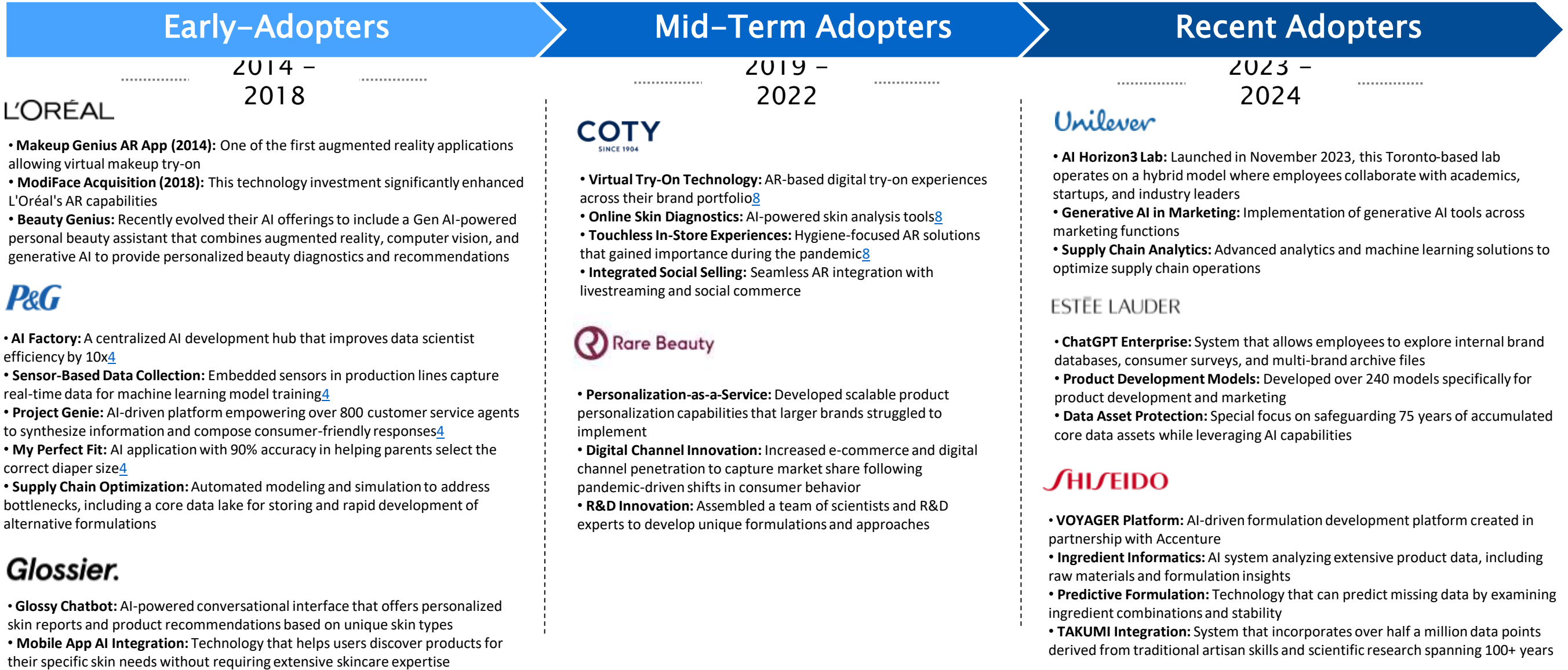
■ L'Oréal ■ P&G ■ Estée Lauder ■ Unilever ■ Shiseido ■ Coty ■ Glossier

Source: <https://www.thebusinessresearchcompany.com/report/ai-in-beauty-and-cosmetics-global-market-report>

Analysis of Market Share Changes Shows Positive Momentum Across All Eight Companies, With Varying Magnitude of Improvement

Dramatic Market Share Gainers	Substantial Market Share Improvers	Moderate Market Share Gainers
• Rare Beauty: Extraordinary 566.67% market share growth, from 0.03% in 2022 to 0.20% in 2023 • Glossier: Impressive 200.00% market share increase, from 0.05% in 2018 to 0.15% in 2023	• Coty: 27.35% market share growth, from 2.34% in 2020 to 2.98% in 2024 • Unilever: 26.98% market share increase, from 29.69% in 2022 to 37.70% in 2024 • P&G: 26.97% market share growth, from 37.52% in 2018 to 47.64% in 2024 • Estée Lauder: 26.96% market share improvement, from 6.75% in 2023 to 8.57% in 2024	• L'Oréal: 9.09% market share growth, from 22.00% in 2014 to 24.00% in 2024 • Shiseido: 4.00% market share increase, from 2.50% in 2023 to 2.60% in 2024

Leading beauty brands are embracing AI at different stages, shaping the industry’s innovation timeline. From early adopters to recent entrants, each phase reflects evolving priorities and capabilities.



Source: <https://www.henrystewartpublications.com/sites/default/files/DSM9.1Howl%E2%80%99Or%C3%A9aladoptednewtechnologiestoscalepersonalisation.pdf>

Glossier’s Expansion Challenge: Balancing Operational Scale and Authentic Brand Identity

While most beauty and cosmetics companies perform well financially, the industry faces a critical challenge: how to scale operations and reach without diluting brand authenticity — a core issue for Glossier, which must implement AI strategically to grow while preserving its community-driven identity.

Company	AI – Focus	Scalability Impact (1–10)	Authenticity Impact (1–10)
Glossier	Community-driven marketing, micro-influencers	6	9
L'Oréal	Generative AI, AR, Beauty Genius	9	8
P&G	Consumer insights, supply chain optimization	8	7
Estée Lauder	Generative AI, ConsumerIQ, Trend Studio	8	8
Unilever	AI Horizon3 Lab, Smart Freezers	9	8
Shiseido	VOYAGER platform, TAKUMI integration	7	8
Coty	Digital Accelerator, AR experiences	7	7
Rare Beauty	TikTok engagement, inclusivity campaigns	8	10

The Scalability–Authenticity Challenge

Beauty brands, especially digital-native ones like Glossier, struggle with balancing growth and maintaining brand authenticity.

- **Core Dilemma:** As these brands scale, they risk weakening the personal, community-driven connections that contributed to their early success.
- **Glossier’s Challenge:** Harvard Business School highlights Glossier’s central issue as "scaling up without losing the brand's unique connection to its audience."
- **Brand Identity at Risk:** For community-built brands, rapid expansion can erode the authenticity that sets them apart.
- **Strategic Need:** Glossier must implement AI thoughtfully to support growth while preserving its original brand values and customer intimacy.

Glossier’s Specific Challenge

- **Glossier’s foundation** is rooted in direct community engagement and close customer relationships.
- **As the brand expanded**, it faced challenges in maintaining its personal, authentic touch.
- **HBS Senior Lecturer Jill Avery** noted the company needed to strategically assess how to integrate paid micro-influencers without damaging its credibility.
- **This situation highlights** the core tension between operational scale and preserving authentic brand identity.

Different AI strategies reveal how beauty brands navigate the trade-off between scaling efficiently and staying true to their brand identity.

Community-Centered AI Approaches

Rare Beauty

- Achieves the best balance between scalability and authenticity (10/10 authenticity, 8/10 scalability).
- Uses AI to enhance TikTok engagement and drive inclusive campaigns.
- Leverages AI to spotlight conversations on mental health, body positivity, and self-acceptance.
- Maintains authenticity while achieving rapid brand growth.

Glossier

- Uses AI to support its community-driven marketing model.
- Analyzes and segments its micro-influencer network for authentic brand representation.
- Maintains its authentic voice by amplifying real community members rather than relying on scaled marketing efforts.
- Focuses on expanding reach without compromising brand identity.

Technology Forward AI Approaches

L'Oréal

- Adopts a technology-forward AI strategy, focusing on generative AI and augmented reality tools.
- Utilizes the GenAI Content Lab to create AI-driven beauty content with brand consistency controls.
- Achieves high scalability (9/10) and strong authenticity preservation (8/10).

P&G

- Leverages AI for consumer insights and supply chain optimization.
- Achieves high scalability (8/10), though with lower authenticity impact (7/10).
- AI Factory boosts data scientist efficiency by 10x.
- Project Genie helps customer service agents deliver consumer-friendly responses through AI-powered synthesis.

Enterprise Integration Approaches

Estée Lauder

- Partnered with Microsoft to develop ConsumerIQ, an AI agent built in Copilot Studio.
- Enables employees to quickly access decades of consumer insights for trend-responsive marketing.
- Achieves a balance of scalability (8/10) and authenticity (8/10).

Unilever

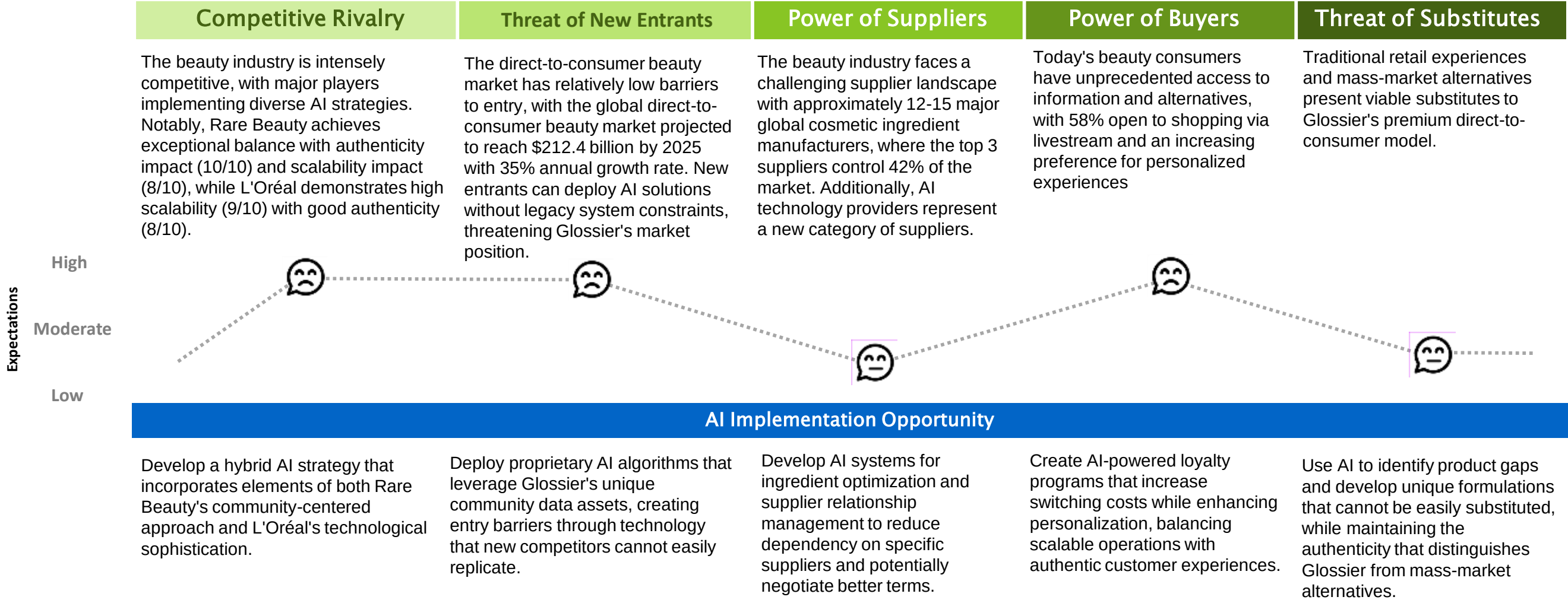
- Runs the AI Horizon3 Lab on a hybrid collaboration model with employees, academics, and startups.
- Focuses on innovation while maintaining high scalability (9/10) and brand authenticity (8/10).

Industry analysis reveals significant variations in AI implementation approaches and their impact on scalability and authenticity

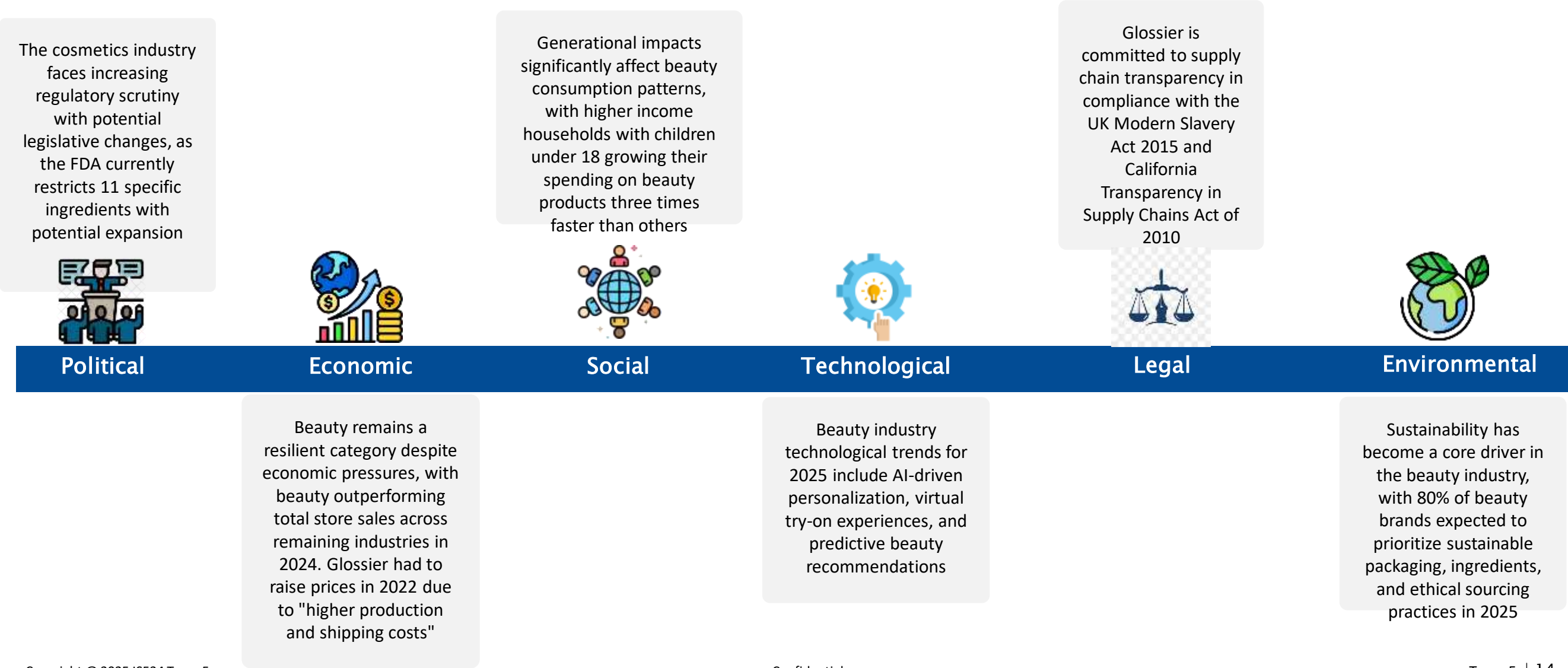
		AI Focus	Scalability Impact (1-10)	Authenticity Impact (1-10)	Key Technology
Company		Community-driven marketing, micro influencers	6	9	Glossy Chatbot, basic analytics
		Generative AI, AR, Beauty Genius	9	8	ModiFace acquisition, GenAI Content Lab.
		TikTok engagement, inclusivity campaigns	8	10	Personalization-as-a Service
		Consumer insights, supply chain optimization	8	7	AI Factory, Project Genie
		Generative AI, ConsumerIQ, Trend Studio	8	8	ChatGPT Enterprise integration
		AI Horizon3 Lab, Smart Freezers	9	8	Hybrid collaborative model with academia

Strategic Landscape Analysis for Glossier Using Analytical Frameworks

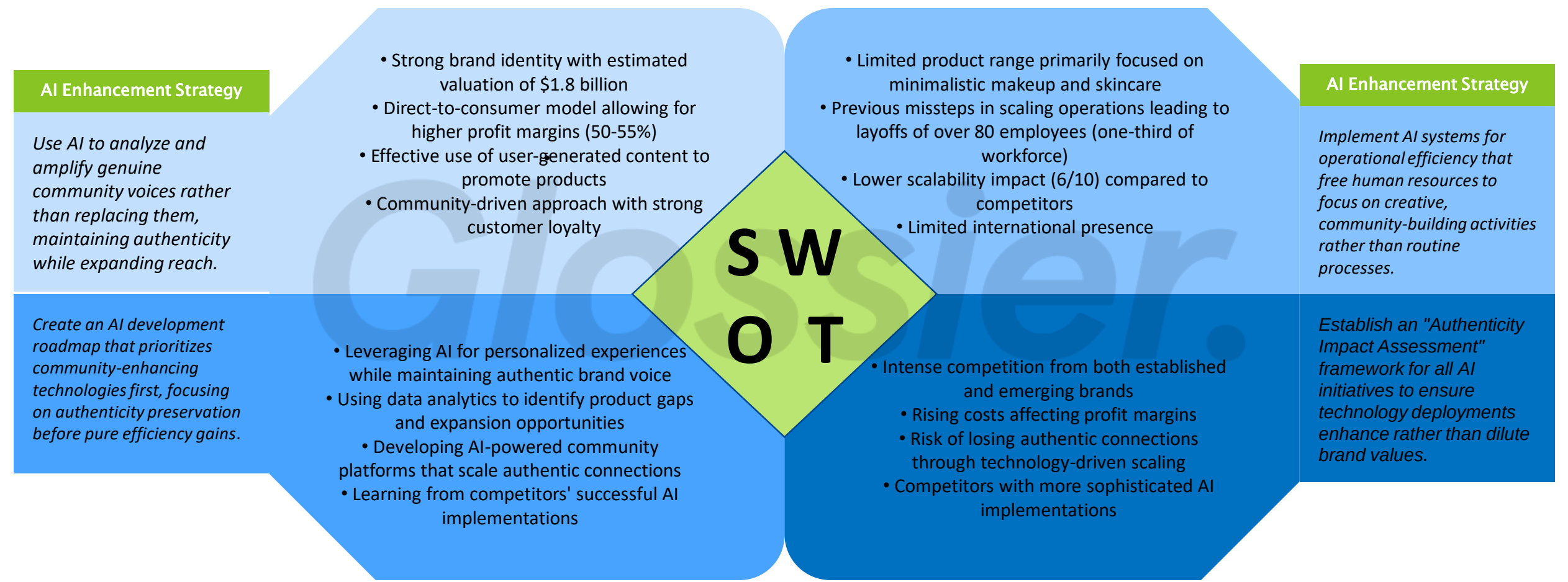
Glossier faces high competitive intensity and pressure across all five forces—operating in a market where 58% of buyers seek personalization and top suppliers control 42% of inputs. With a scalability score of just 6/10, it must navigate these challenges while preserving its community-driven identity.



Glossier operates in a rapidly evolving beauty landscape shaped by regulatory shifts, rising costs, changing consumer demographics, and sustainability demands—requiring thoughtful adaptation to maintain its brand integrity and market relevance.



With authenticity as its core strength, Glossier's growth depends on thoughtful AI adoption that strengthens rather than replaces community engagement.



The New Beauty Consumer: Balancing Technology Adoption and Human Connection

From in-store consultants to AI-powered recommendations, the beauty journey is now more personalized and tech-driven. Consumers seek virtual try-ons, tailored advice, and a balance between technology and human input

Evolution of Customer Expectations

- The beauty customer journey has transformed significantly with the rise of AI technologies.
- Traditionally, consumers depended on **in-store assistance and beauty consultants** for personalized product advice.
- The shift to **online shopping** introduced reliance on **customer reviews**, which were often inconsistent and subjective.
- **AI now addresses these limitations** by delivering **personalized recommendations** based on data-driven analysis of individual traits and preferences.

Key Customer Behaviors: Modern beauty consumers demonstrate several distinctive behaviors when interacting with AI-powered beauty solutions



Desire for Personalization

According to market research, customers are three times more likely to purchase cosmetic products if they can digitally try them on before purchasing



Engagement with Virtual Try-On

L'Oréal reported a 52% increase in e-commerce sales in 2019 due to AR and AI technologies adoption, demonstrating the significant impact of interactive technologies on purchase behavior



Trust in AI Generated Recommendations

Consumers are increasingly trusting AI-powered skin assessments and recommendations, particularly when these tools provide detailed analysis of metrics like skin texture, hyperpigmentation, and wrinkles



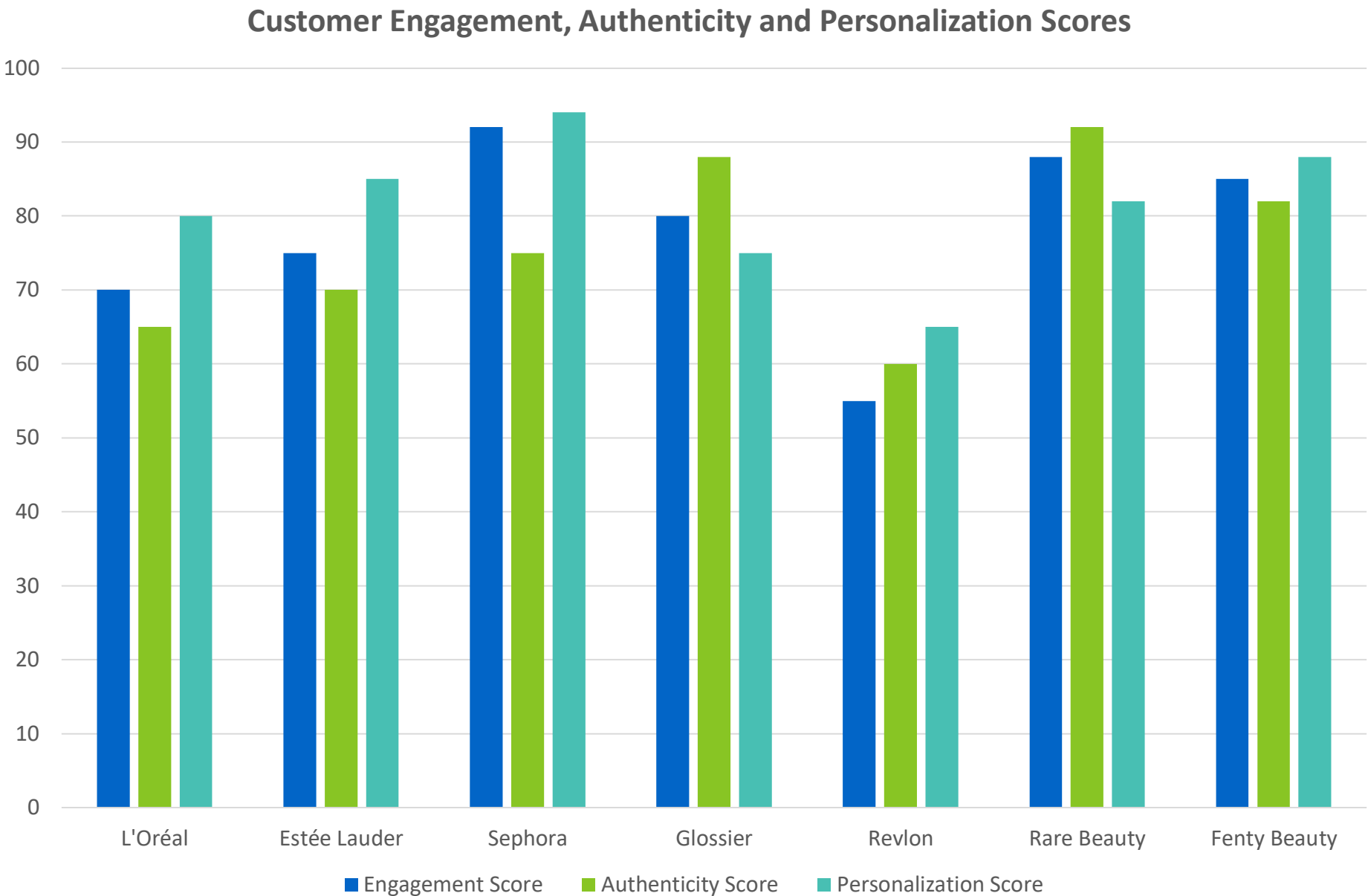
Community-Driven Input

Despite the rise of AI, customers still value human connection and community input in their beauty decisions. The most successful brands are those that balance technological innovation with authentic community engagement

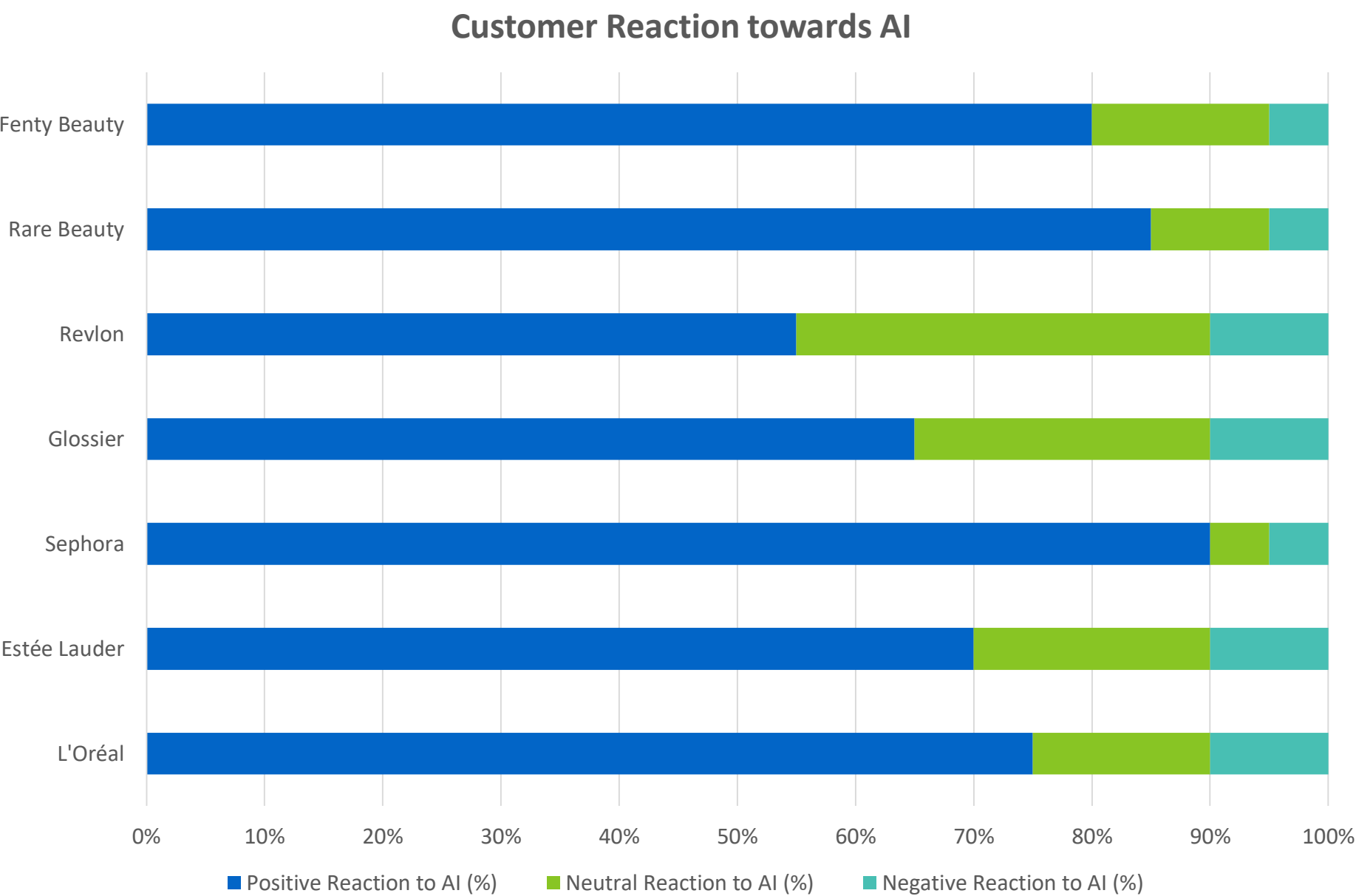
Customer interaction data shows Glossier ranks high in authenticity but lags in personalization, signaling a clear area for strategic AI enhancement in boosting personalization

Key Insights

- Many brands, especially Sephora and Fenty Beauty, showcase high personalization scores, indicating a focus on delivering tailored experiences to customers, potentially through loyalty programs and online AI integration.
- The level of engagement is variable across brands, but those excelling (Sephora, Fenty, Rare) tend to perform well in at least one of the other categories, suggesting customer retention is linked to overall brand strategy.
- Glossier excels in authenticity, ranking high among competitors. However, its personalization score is comparatively lower, indicating a potential area for improvement.
- There is a potential for Glossier to improve personalized experiences. Based on the document this can be done through adopting AI technology tools like virtual try-ons to better meet the user needs and improve engagement with the website.



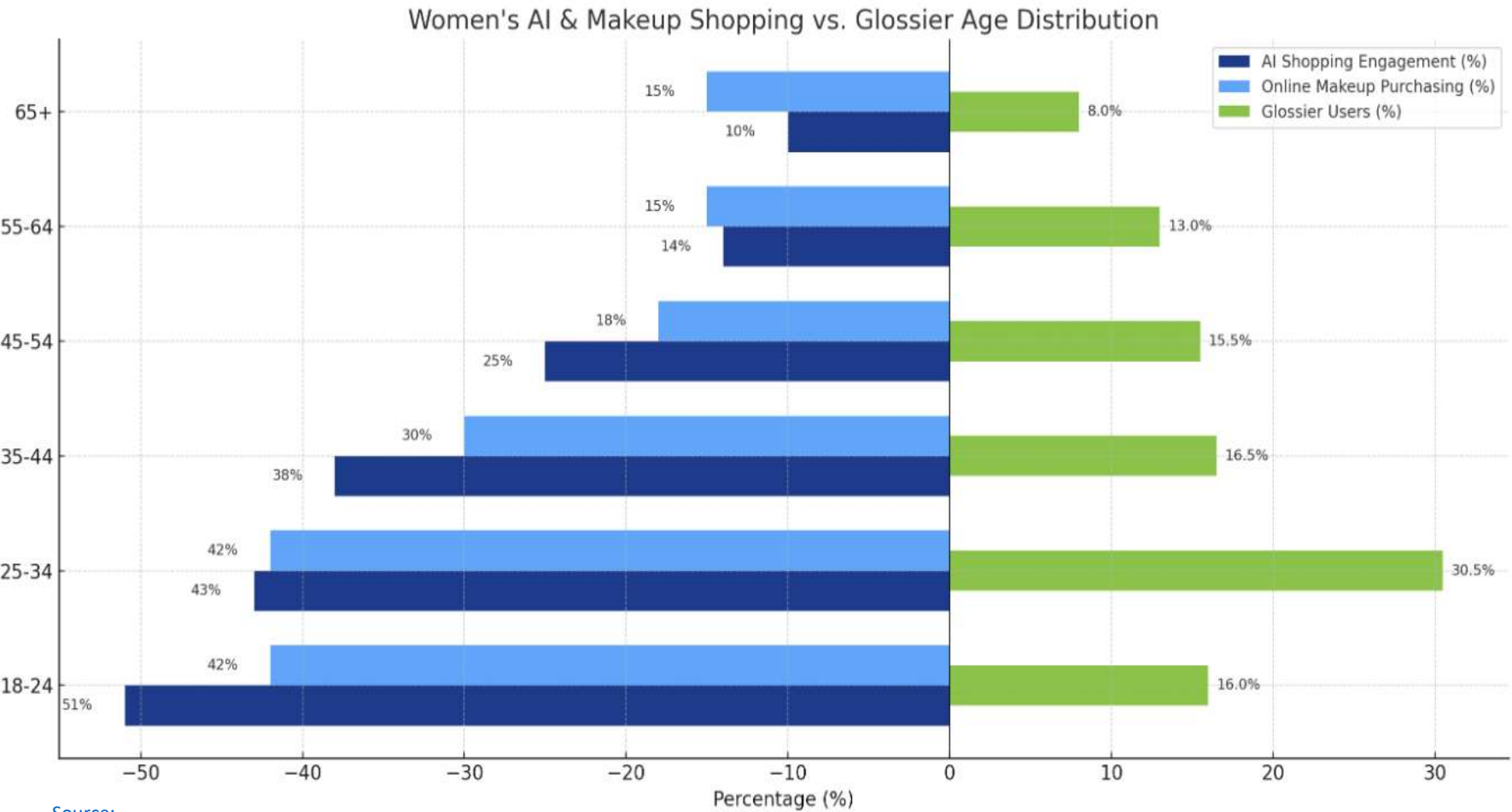
While most beauty brands receive overwhelmingly positive responses to AI, Glossier's audience shows more caution — highlighting the need for value-aligned, authenticity-preserving AI integration.



Key Insights

- Overall, there's a strong positive reaction towards AI implementation in the beauty industry across most brands. Most have 70-90% of responses in the 'positive reaction' range.
- Despite the rapid influx and adaptation of AI, there are few negative reactions. People might be scared but they are not strongly opinionated.
- While still positive, Glossier's customer base has a slightly lower positive reaction to AI compared to Fenty, Rare, and Sephora. This could be because its audience values authenticity and realness, potentially viewing AI with some skepticism.
- To improve, Glossier could consider incorporating AI in ways that enhance, rather than replace, its core values. AI integration can occur behind-the-scenes in its user review platform or customer support platform instead of changing the brand.

AI Beauty Tech adoption is highest in the Women’s age segment of 18–44 which shows strong digital engagement, frequent online purchases, and openness to AI-powered shopping experiences



Source:
<https://news.cognizant.com/2025-01-17-Cognizant-Study-Shows-Consumers-Who-Embrace-AI-Could-Drive-4-4-Trillion-in-Spending-Over-Five-Years>
<https://brandequity.economictimes.indiatimes.com/news/research/48-of-online-shoppers-are-women-report/113033898>
<https://www.emarketer.com/content/how-us-consumers-shop-beauty-across-age-groups>

Age Demographics Insights

65+: Low adoption of AI (Voice assistant, auto recommendation). Age group prefers shopping in store.

55-64: Growing adoption; cautious but curious about AI-led product suggestions. Age group prefers shopping in store.

45-54: Moderate interaction, often through virtual assistants. Has a preference for in-store shopping.

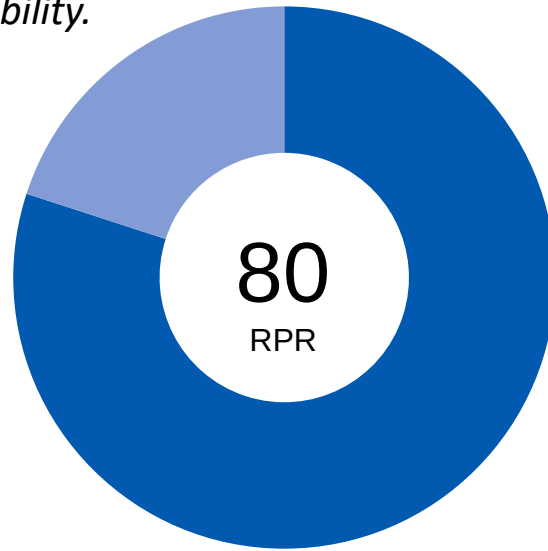
35-44: High awareness; use AI for personalization and price comparison. Women shop for beauty products online monthly.

25-34: High adoption Daily AI users. 42% shop for beauty products online monthly.

18-24: Use AI for product discovery, try-ons, and recommendations. High frequency of online beauty purchases, with 21% buying daily.

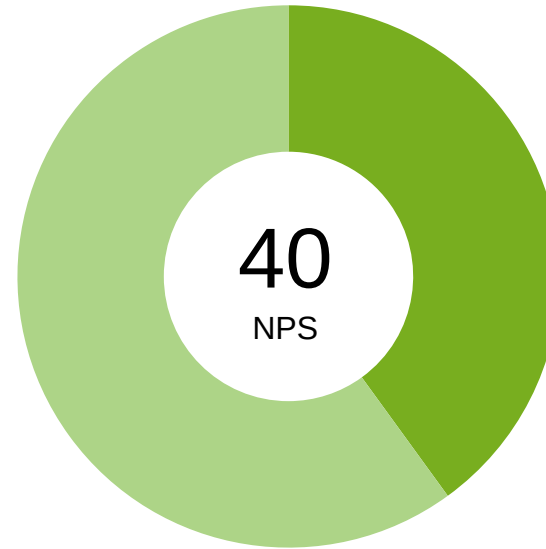
Glossier's strong Repeat Purchase Rate and Referral Contribution reflect high customer loyalty, while a moderate NPS highlights room to grow brand advocacy.

Loyalty metrics are a set of measurements used by businesses to understand and quantify customer allegiance to a brand, product, or service. They provide insights into customer behavior, satisfaction, and likelihood of continued engagement, which are crucial for sustainable growth and profitability.



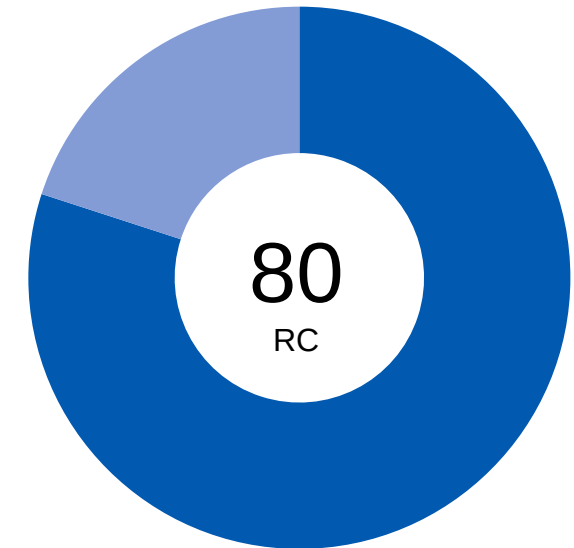
Repeat Purchase Rate

This metric measures the percentage of customers who make more than one purchase from a brand within a specific time frame. A high repeat purchase rate indicates strong customer satisfaction, brand loyalty, and effective product offerings. Focusing on increasing this rate can lead to sustained revenue growth.



Net Promoter Score

NPS quantifies customer loyalty by asking customers how likely they are to recommend a brand on a scale of 0-10. It's a simple yet powerful indicator of brand health, with a higher score reflecting a greater proportion of enthusiastic brand advocates.



Referral Contribution

This metric tracks the percentage of new customers acquired through referrals from existing customers. A high referral contribution signifies strong brand advocacy and customer trust, indicating that the brand is delivering value and creating positive customer experiences.

Real-Time Peer Group Consumer Survey on AI in Cosmetics Purchase

Survey reveals evolving buyer behavior, virtual try-on experiences, and expectations from AI-powered cosmetic technologies

Beauty Product Preferences & AI Virtual Try-On Experience Survey

HRI 📱
The form is aimed to gather insights from customers who use beauty products and their attitudes toward AI virtual technology in the beauty industry. Your responses will help us better understand your shopping habits, brand preferences, and experiences with virtual try-on tools. Your answers are confidential and will be used for research purposes only.

r143@illinois.edu [Switch account](#)
Not shared

* Indicates required question

How often do you purchase beauty products?

- ☐ Once a week
- ☐ Once a month
- ☒ Every 3-6 months
- ☐ Rarely

What beauty brands do you usually purchase? (Select all that apply)

- ☐ Sephora
- ☒ L'Oréal
- ☐ Estée Lauder
- ☐ Glossier
- ☐ MAC
- ☐ Fenty Beauty
- ☒ Dior
- ☐ Chanel
- ☐ Maybelline
- ☐ NYX

What would influence your purchase decisions? (Select all that apply)

- ☒ Online reviews
- ☐ Social media influencers
- ☒ Friends and family recommendations
- ☐ In-store recommendations

How satisfied were you with the virtual try-on experience?

- ☐ Very satisfied
- ☐ Somewhat satisfied
- ☒ Neutral
- ☐ Not satisfied
- ☐ Not applicable

Do you trust AI virtual try-on results when purchasing?

- ☐ Yes, completely
- ☒ Somewhat, but I double check in store
- ☐ No

Do you feel AI virtual try-on reduces the need for product returns?

- ☐ Yes
- ☐ No
- ☒ Not sure

What do you think AI beauty tech needs to improve?

- ☐ Color accuracy
- ☐ Skin tone detection
- ☐ Realism of application
- ☐ User interface
- ☒ Other: Precision

Submit Clear form

Do you try on beauty products before purchasing?

- ☐ Yes, always
- ☒ Sometimes
- ☐ No, I buy directly

What beauty products have you tried with AI virtual try-on? (Select all that apply)

- ☒ Lipstick
- ☐ Foundation
- ☐ Eyeshadow
- ☐ Eyeliner
- ☐ Blush
- ☐ Mascara
- ☒ Hair color
- ☐ Not applicable

Would you be more likely to buy a product after using AI try-on?

- ☒ Yes
- ☐ Maybe
- ☐ No

Do you think AI virtual try-on is better than in-store testers?

- ☒ Yes, AI is more hygienic and convenient
- ☐ No, in-store testing is more reliable
- ☐ Both have pros and cons

Do you see AI as a positive innovation in the beauty industry?

- ☒ Yes, it improves the shopping experience
- ☐ Neutral
- ☐ No, it's unnecessary

What concerns do you have about AI in beauty? (Select all that apply)

- ☒ Data privacy
- ☒ Accuracy of recommendations
- ☐ AI replacing human expertise
- ☐ None

Customer Personas Based on the Survey



Emma

Tech-Savvy Early Adopter

Demographics: 28, urban professional, high disposable income
Purchasing Habits: Buys beauty products once a month; prefers luxury brands like Dior, Chanel, and Estée Lauder.
AI Interaction: Uses AI virtual try-on tools, very satisfied, and trusts results completely. Believes AI is more reliable and convenient than in-store testers.
Influences: Relies on online reviews and social media influencers.
Goals: Seamless, innovative shopping experiences.
Frustrations: Limited realism in current AI applications.
Quote: “AI saves me time and lets me experiment with looks I’d never try in-store.”



Amara

Cautious Hybrid Shopper

Demographics: 35, suburban parent, mid-range budget
Purchasing Habits: Buys every 3-6 months; mixes drugstore (Maybelline, L'Oréal) and premium brands (MAC, Sephora).
AI Interaction: Uses AI try-on but double-checks in-store. Somewhat satisfied but worries about color accuracy and skin tone detection.
Influences: Trusts friends/family recommendations and in-store advice.
Goals: Balance convenience with reliability.
Frustrations: “AI tools don’t capture lighting or undertones well.”
Quote: “I’ll use AI to narrow options, but I need to see it in person.”



Linda

Traditional In-Store Loyalist

Demographics: 45, rural resident, budget-conscious

Purchasing Habits: Rarely buys beauty products; sticks to affordable brands like NYX and Maybelline.

AI Interaction: Avoids AI tools; prefers in-store testers. Distrusts virtual results and cites data privacy concerns.

Influences: Relies on in-store recommendations and online reviews.

Goals: Assurance of product quality before purchase.

Frustrations: “AI can’t replicate the feel of a product or how it oxidizes.”

Quote: “If I can’t touch it, I won’t buy it.”



Dr. Priya

Skeptical Innovator

Demographics: 40, dermatologist, tech-curious but critical

Purchasing Habits: Buys monthly; prefers science-backed brands (Estée Lauder, MAC).

AI Interaction: Uses AI tools but not satisfied. Highlights flaws like poor skin tone detection and lack of realism.

Influences: Prioritizes online reviews and professional insights.

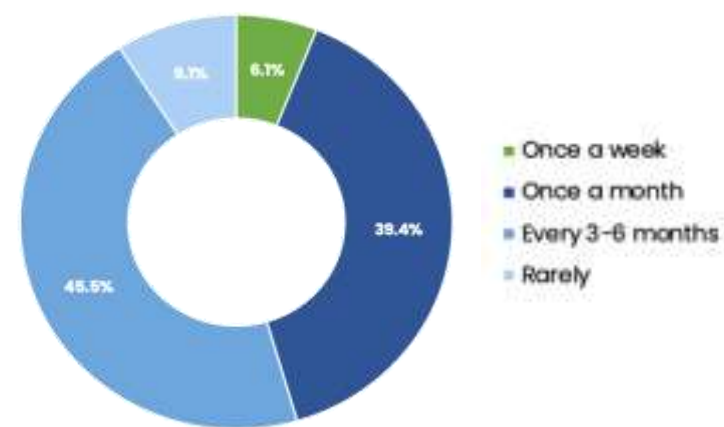
Goals: Precision in shade matching and product performance.

Frustrations: “AI ignores formulation and finish—key factors for my clients.”

Quote: “AI has potential, but it’s not there yet for professionals.”

The Real time survey of customer expectations shows most beauty consumers buy products every few months, while AI virtual try-on tools are widely used, highlighting growing comfort with tech-driven beauty shopping

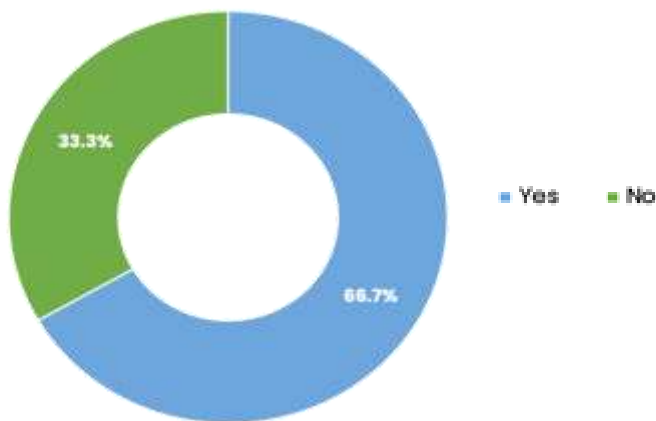
Frequency of Beauty Product Purchases (Approx. %)



Customers Buying Habits Insights

- Most users follow a maintenance-driven buying pattern, with **45.5%** restocking every 3-6 months. This seasonal purchasing behavior suggests that brands can maximize engagement by aligning promotions with these cycles.
- **39.4%** of users purchase beauty products monthly, representing a loyal segment ideal for subscription services and rewards programs.
- A smaller group, **6.1%** purchases weekly. **9.1%** who rarely purchase present an opportunity for reactivation campaigns through targeted promotions

AI Virtual Try-On Usage for Beauty Products (Approx. %)

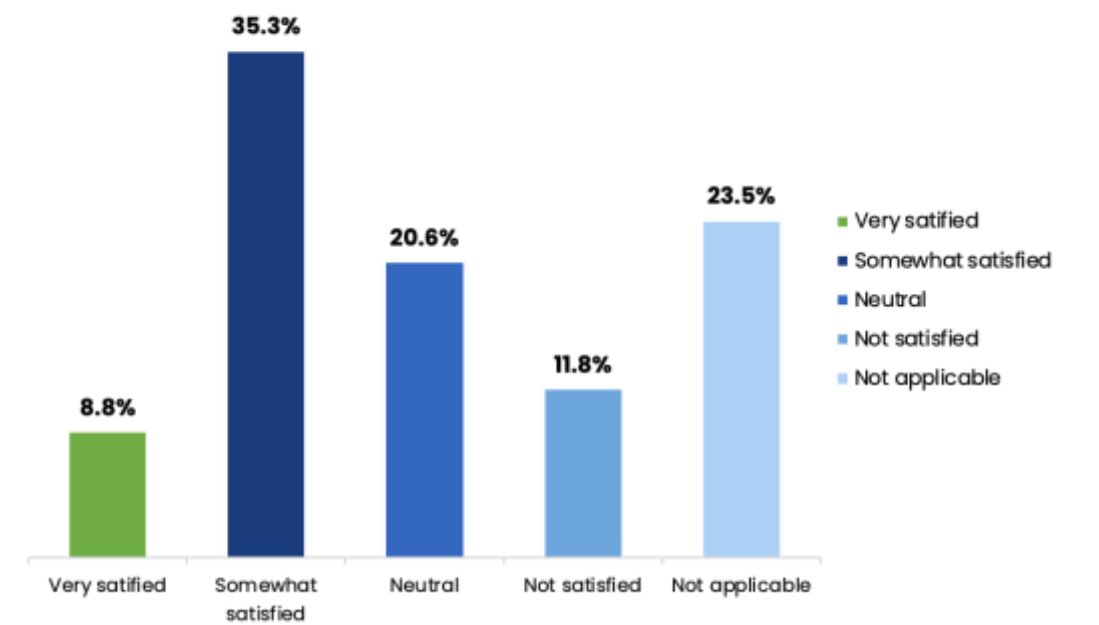


Analysis of AI Virtual Try-On Usage Experience

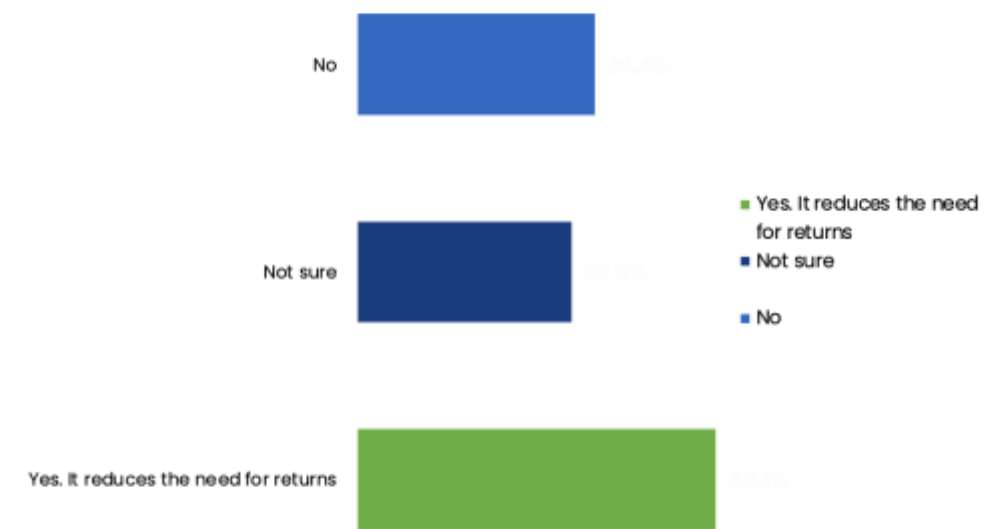
- A large majority has already experienced virtual try-on, proving AI integration is no longer optional but expected in beauty tech.
- This widespread adoption highlights the importance of AI integration aiming to remain competitive and meet consumer demands for personalized, tech-driven shopping experiences.
- However, with **33.3%** of users still unfamiliar with these tools, there remains significant potential for market expansion.
- Targeted outreach, especially toward older or less tech-savvy consumers can help bridge this gap and further increase usage rates.

The results highlight a promising outlook for AI virtual try-on tools, with many users reporting satisfaction and recognizing their value in enhancing the shopping experience.

User Satisfaction with Virtual Try-On Experience



Impact of AI Try-On on Product Returns

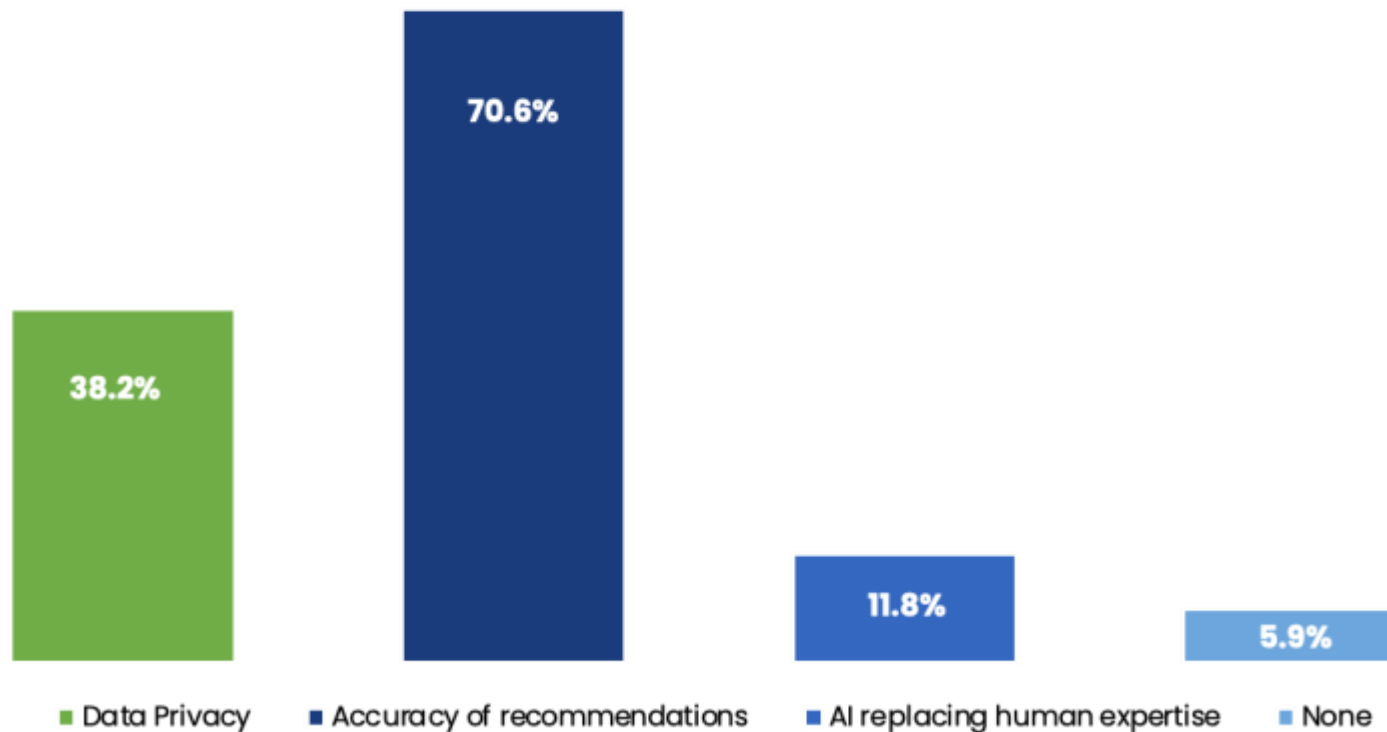


Customers Attitudes towards Virtual Try-On Experience

- **8.8%** of users are highly satisfied with virtual try-on experience and **35.3%** of users found the AI virtual try-on helpful but felt there is potential for improvement.
- **44.1%** of users believe that virtual try-on helps reduce product returns, reflecting growing trust in the technology’s ability to improve purchase confidence and minimize mismatches.
- These findings emphasize the increasing role of AI try-on features in driving positive user experiences and supporting smarter beauty product decisions.

Accuracy of AI recommendations is the biggest concern among users, followed by data privacy risks. Fewer users worry about AI replacing human expertise, while a small group reports no concerns at all

Concerns Regarding AI Technology in Beauty Industry



Main Concerns about AI techs

- **70.6%** of users expressing doubts about how well AI can match beauty products to individual needs.
- Data privacy follows as the second major concern, cited by **38.2%** of users who are wary of sharing personal information.
- A smaller portion, **11.8%**, worries about AI replacing human expertise, while only **5.9%** report having no concerns.
- These results highlight the need for brands to prioritize transparency, improve AI accuracy, and balance technology with human input to build user trust.

Evaluating Glossier's Technology Backbone: A Snapshot of Existing AI Capabilities and Infrastructure

Glossier's technology infrastructure centers on its e-commerce platform, which was migrated to Shopify Plus in late 2022. This provides a stable foundation for AI integration with established APIs and development environments.

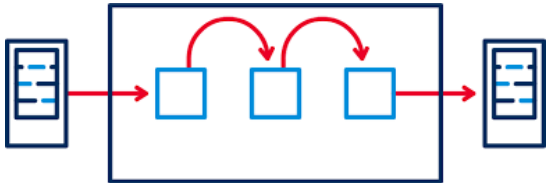
Key Components				
E-Commerce Platform Shopify Plus with custom integrations	Customer Data Platform Shopify Plus with custom integrations	Content Management System Shopify Plus with custom integrations	Supply Chain Management Inventory and Order Management Systems	Customer Service Platform Support for customer inquiries and feedback

Current AI Implementations



Glossy Chatbot

An AI-powered conversational interface offering personalized skin reports and product recommendations based on unique skin types



Cross-Domain Learning

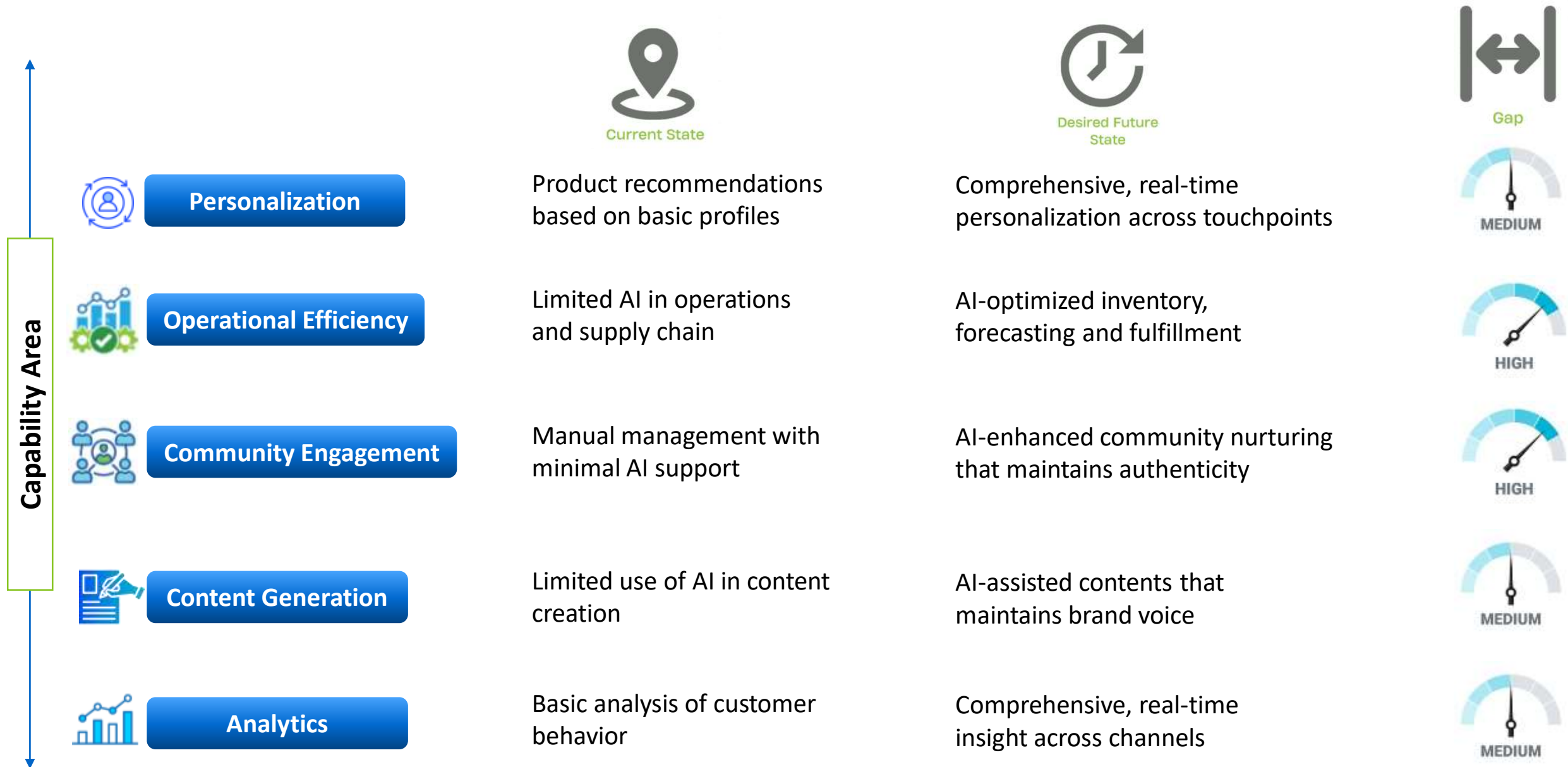
Machine Learning systems that track customer identities across channels and identify profitable patterns in cross-platform behavior



Content Analysis

Basic analysis of user-generated content to inform product development

Identifying Capability Gaps in AI Adoption: A Comparative Snapshot of Glossier's Current State vs. Industry-Leading Practices Across Key Business Areas



Strategic Insights from Competitor AI Adoption and Customer Analysis: Lessons for Glossier’s Scalable and Authentic Growth

Considerations for Glossier to overcome its core challenge in Balancing Scalability and Brand Authenticity			
Timing Affects Strategy Early AI adopters had to pioneer approaches, while recent adopters have implemented more integrated, enterprise-wide solutions informed by industry learnings.	Community Integration Is Key The most successful authenticity preservation occurs when AI enhances rather than replaces community connections. Rare Beauty and Glossier exemplify this approach.	Optimal Balance Is Achievable Companies like Estée Lauder demonstrate that with careful implementation, AI can support both scalability and authenticity simultaneously.	Implementation Maturity Matters Companies with longer AI implementation periods generally show more sophisticated integration between authentic brand values and scalable operations.

Glossier's customers value the brand primarily for its authentic community connections and personalized approach. Analysis of customer engagement data reveals		
Loyalty to Brand Voice Customers demonstrate strong attachment to Glossier's authentic communication style and community-driven approach.	Appreciation for Personalization The Glossy Chatbot and personalized recommendations have been well-received, with positive customer feedback on the relevance of suggestions.	Concerns About Scaling Some longtime customers express concern about potential loss of the "personal touch" as the brand grows, highlighting the importance of maintaining authenticity through scaling.

From the analysis, The suggestion for Glossier is that maintaining its community-centered approach while expanding its AI capabilities in consumer insights and personalization could help address its core challenge. By learning from competitors like Rare Beauty that have achieved higher scalability while maintaining authenticity, Glossier could potentially evolve its AI strategy to better support growth without compromising its authentic brand connection.

The most promising path forward for Glossier is to develop AI solutions that specifically enhance human connections rather than automating them away, preserving the authentic community engagement that has been central to the brand's success while enabling the operational efficiency needed for continued growth.

Prioritized Technology Strategy: Strengthen Community Engagement and Operational Efficiency

Based on comprehensive evaluation, the recommended technology solution is a phased implementation prioritizing the Hybrid Community Management System, followed by the Community Intelligence Platform, and then AI-Powered Supply Chain Optimization.

Technical Architecture		
Core System	Supporting System A	Supporting System B
Hybrid Community Management System - AI-powered conversation routing and prioritization - Response suggestion with human review - Community member segmentation and personalization - Engagement analytics dashboard	Community Intelligence Platform - Integration with social listening tools - Trend identification and forecasting - Micro-community detection - Sentiment analysis across channels	Supply Chain Intelligence - Demand forecasting - Inventory optimization - Fulfillment efficiency - Distribution network optimization

Technology Selection Criteria					
 Alignment with Brand Values	 Technical Maturity	 Integration Capabilities	 Transparency	 SCALABILITY	 Ethical Considerations

Risks and Mitigation Strategies: Safeguarding Glossier’s AI–Driven Future from End to End

Technical Vulnerabilities: Ensuring Data Quality, Scalability, and System Integrity

Technical Risks			
Risk	Impact	Probability	Mitigation Strategy
Data quality issues affecting AI performance	High	Medium	Implement robust data validation and cleaning processes; establish data quality metrics and monitoring
Integrating challenges with existing systems	Medium	High	Conduct thorough technical discovery
Performance degradation at scale	High	Medium	Implement performance monitoring; design for scalability; conduct load testing; establish performance bench marks
AI model drift over time	Medium	High	Establish model monitoring framework; Implement regular retraining schedule; develop version control for models
Security vulnerabilities	High	Low	Conduct security assessment; Implement encryption and access controls; Regular security audits

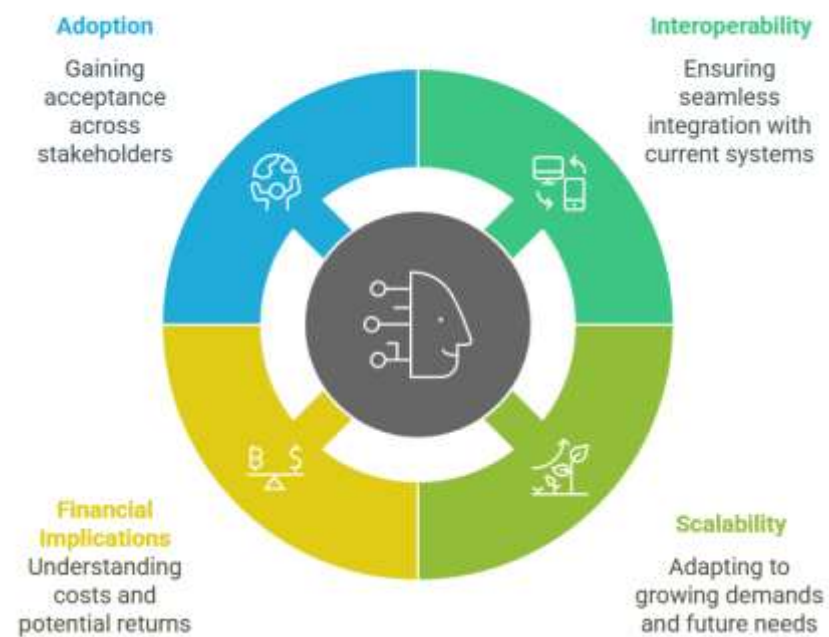
Brand, Budget, and Compliance: Preserving Authenticity While Driving Innovation

Business Risks			
Risk	Impact	Probability	Mitigation Strategy
Perceived loss of brand authenticity	Very High	Medium	Transparent communication; maintain human touchpoints; “AI-enhanced not AI-replaced” messaging
Staff resistance to AI adoption	Medium	Medium	Comprehensive change management; staff training; emphasize AI as augmentation not replacement
Implementation costs exceeding budget	Medium	Medium	Phased approach with clear checkpoints; contingency budget; regular cost reviews
Competitive AI implementations outpacing Glossier	High	Medium	Regular competitive analysis; flexible implementation plan; strategic partnerships
Regulatory compliance issues	High	Low	Regular legal review; privacy impact assessments; ethics review board

Holistic Risk Management: Balancing Technical and Business Priorities for Sustainable AI in Customer Experience

Customer Experience Risks			
Risk	Impact	Probability	Mitigation Strategy
Negative customer reaction to AI implementation	High	Medium	Transparent communication; opt-in features; feedback mechanisms; A/B testing before full rollout
Inconsistent AI performance across customer segments	Medium	Medium	Diverse training data; performance monitoring by segment; bias testing and mitigation
Privacy concerns from customers	High	Medium	Clear privacy policies; transparent data usage; data minimization principles; consent mechanisms
Feature adoption below targets	Medium	Medium	User education; Intuitive design; progressive disclosure of AI features; highlighting tangible benefits
Unintended bias in AI recommendations	High	Low	Bias testing across demographic groups; diverse training data; human oversight; regular audits

Glossier's AI Implementation Risks

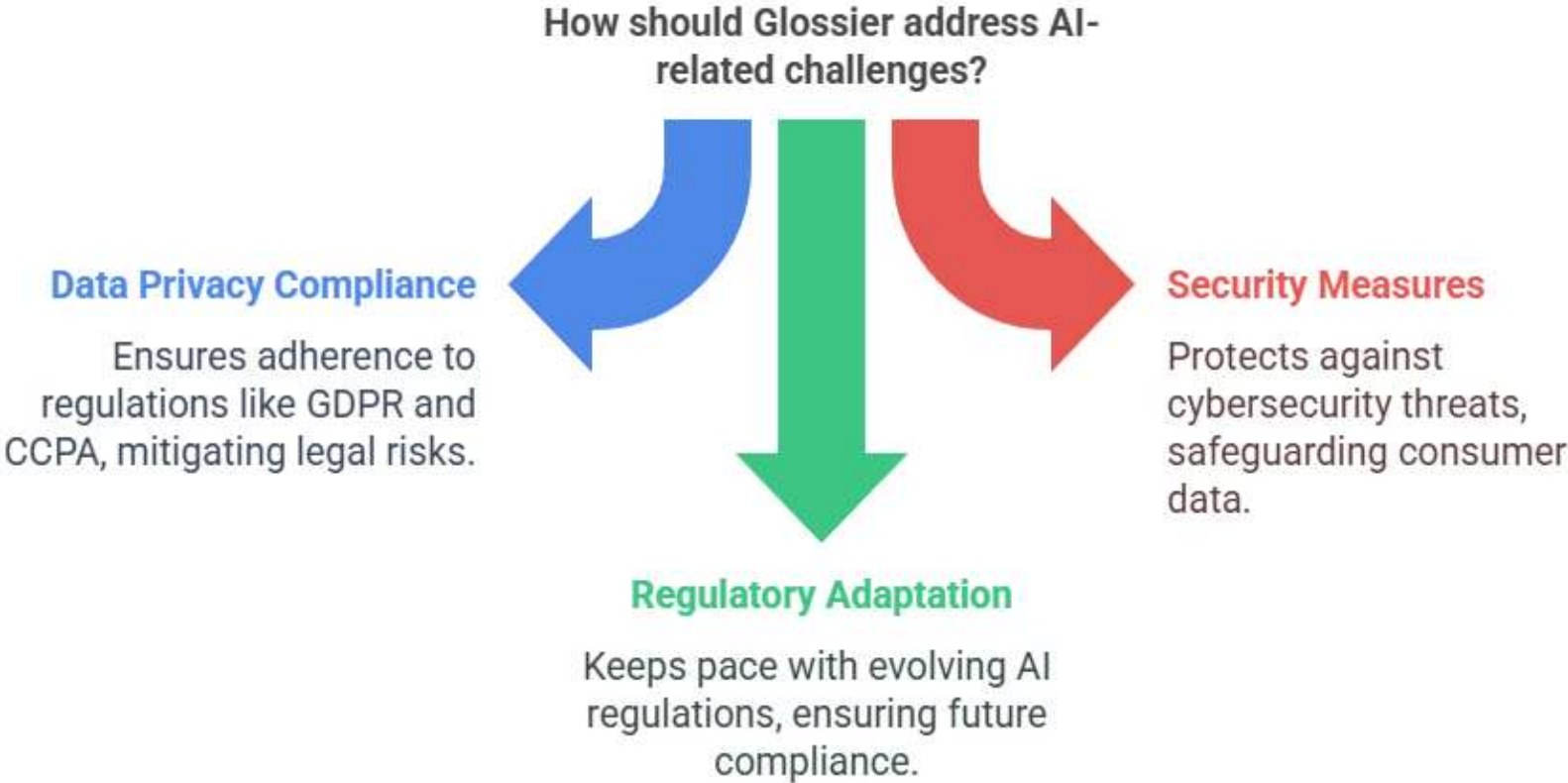


As Glossier explores the adoption of advanced AI technologies to enhance personalization, customer experience, and operational efficiency, it must also navigate a range of strategic and technical risks.

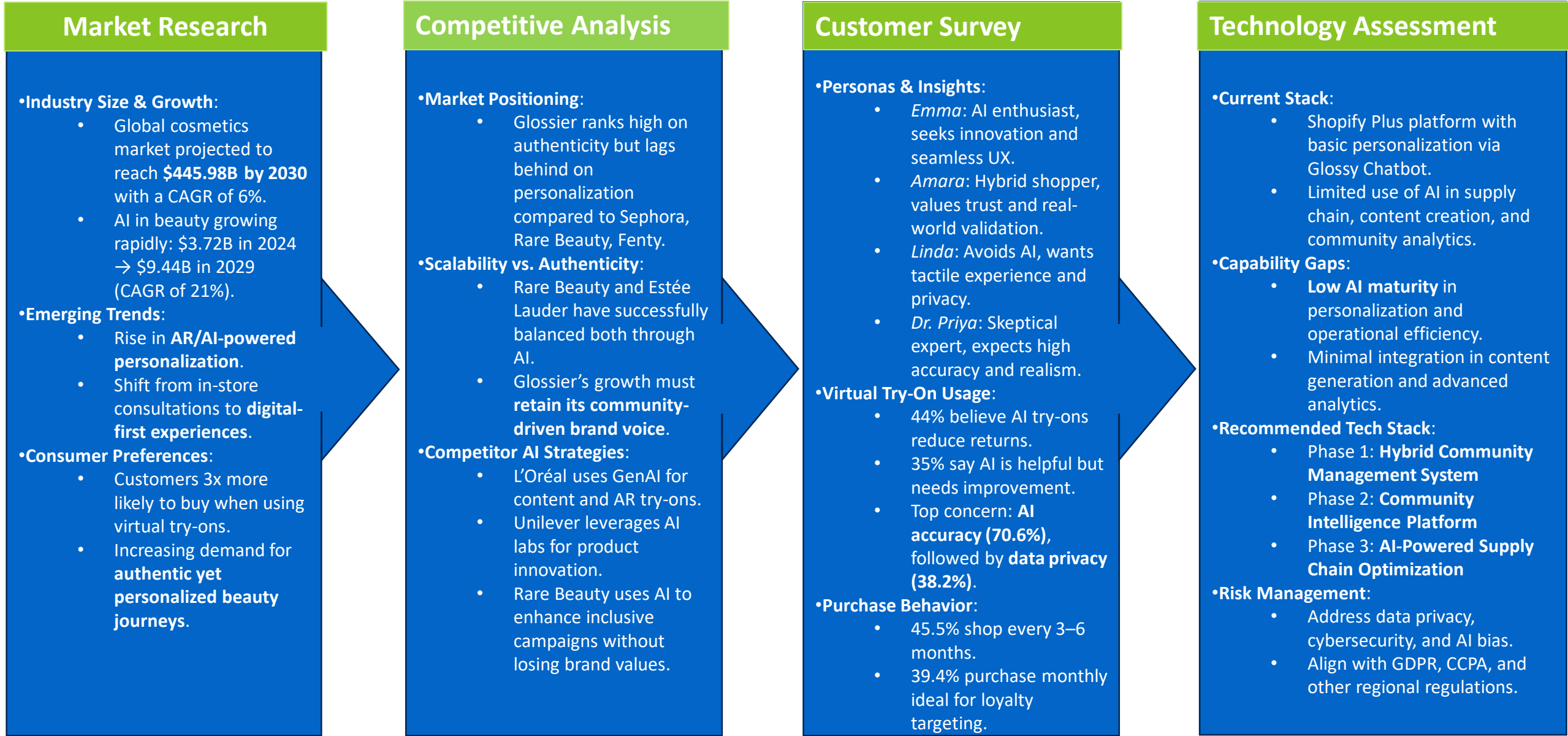
Safeguarding AI Initiatives: Navigating Privacy, Regulation and Cybersecurity Risks

Key Takeaways

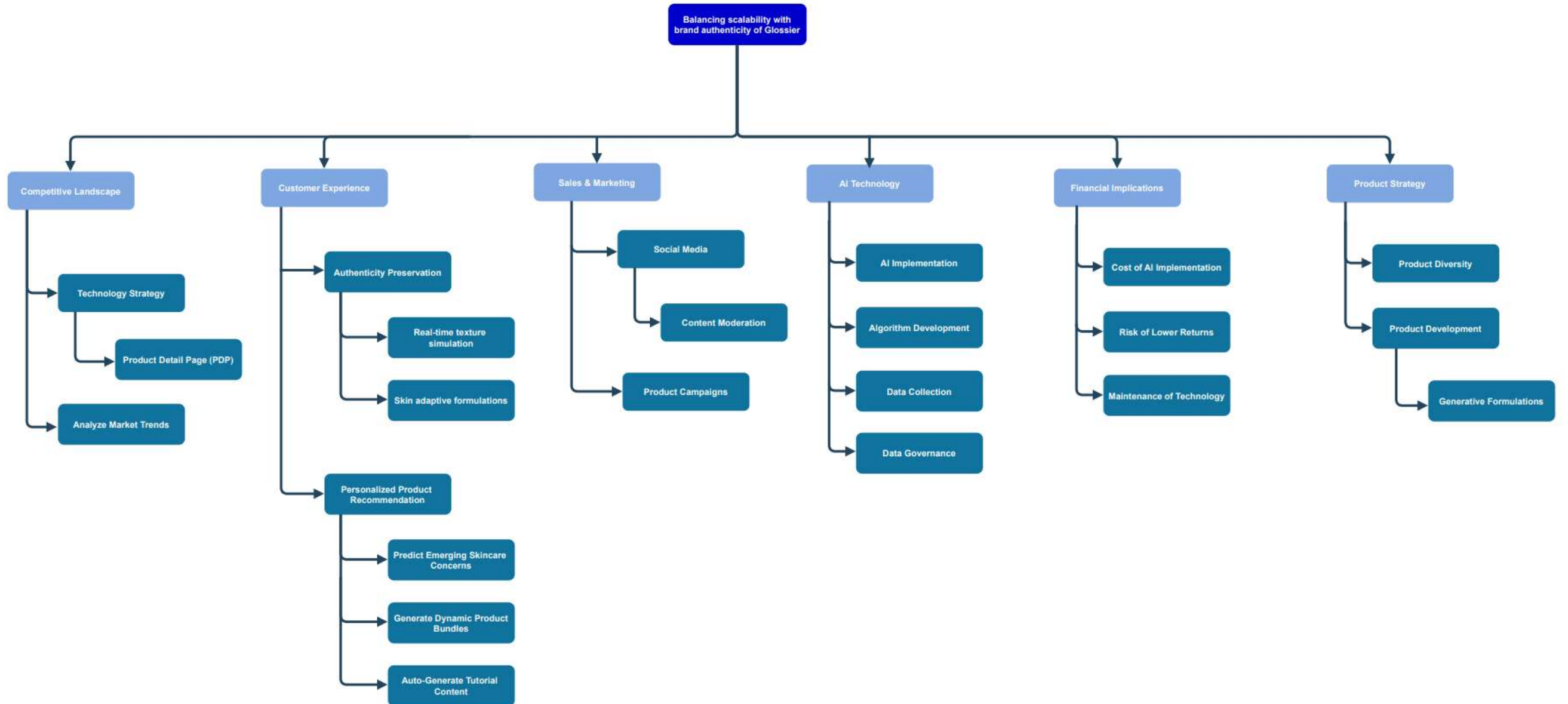
- **Data Privacy Compliance:** The extensive data collection required for personalized AI-driven recommendations increases risks related to privacy compliance. Glossier must clearly define and communicate data handling policies to customers, ensuring full alignment with regulations like GDPR and CCPA.
- **Regulatory Adoption:** The rapidly evolving AI regulatory landscape necessitates Glossier's continuous monitoring and adjustment of practices. Ensuring AI processes remain transparent, ethical, and compliant with emerging laws protects the company from future regulatory risks.
- **Security Measures:** Increased AI implementation amplifies cybersecurity risks, particularly around consumer data protection. Implementing robust cybersecurity measures and continuously updating security protocols will safeguard Glossier's systems against potential breaches and preserve customer trust.



Appendix



Issue Tree



The Need for Real-Time Survey Among Peer Groups



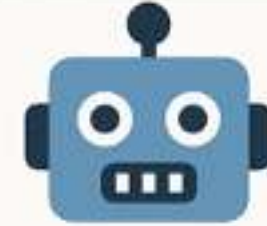
Beauty Purchase Behavior

- Frequency and brand preferences help segment users by engagement and loyalty.
- Identifies leading beauty brands and purchase triggers.



Try-On & AI Experience

- Captures how users interact with beauty products (in-store vs. virtual).
- Measures satisfaction with AI virtual try-on tools.
- Checks trust level in AI suggestions and behavior post-try-on.



AI Attitude & Perception

- Explores willingness to adopt AI for decision-making.
- Evaluates whether AI try-on tools help lower product return rates.
- Identifies perceived strengths and weaknesses.

Bibliography

- <https://ppl-ai-code-interpreter-files.s3.amazonaws.com/web/direct-files/30793844/3fe7a4f1027145cc-a331-b047850da3e6/0/3ebd4907.png>
- <https://d3.harvard.edu/platform-rctom/submission/beauty-brand-with-droves-of-data-how-glossier-employs-machine-learning/>
- <https://www.linkedin.com/pulse/ai-technology-adoption-cosmetic-industry-review-dr-simi-sugathan-ro29c>
- <https://www.dkshdiscover.com/en/articles/hyper-personalization-and-ai-driven-beauty-the-future-of-personal-care/hyperpersonalization>
- <https://ppl-ai-code-interpreter-files.s3.amazonaws.com/web/direct-files/30793844/47bd7ccb-9cb7496283f2-a0a5865b3632/0/5f38abf5.png>
- https://pplx-res.cloudinary.com/image/upload/v1742756832/user_uploads/cYTTXMeIpHXghFY/image.jpg
- https://pplx-res.cloudinary.com/image/upload/v1742756840/user_uploads/kfLlXGggfVqdlv/image.jpg
- <https://beautymatter.com/articles/how-ai-and-data-are-redefining-personalized-experiences>
- <https://gripsintelligence.com/insights/retailers/glossier.com>
- <https://research.contrary.com/company/glossier>
- <https://blog.osum.com/glossier-market-share/>
- <https://www.millionairematch.com/dating-tips/top-10-cosmetic-brands-in-2025>
- <https://canvasbusinessmodel.com/blogs/competitors/glossier-competitive-landscape>
- <https://www.fastcompany.com/90849720/glossiers-sephora-launch-is-part-of-a-bigger-transformation>
- <https://www.42signals.com/blog/rise-of-glossier-from-beauty-blog-to-brand/>
- [https://www.reddit.com/r/glossier/comments/wrxcz0/probably an unpopular opinion but i think a lot/](https://www.reddit.com/r/glossier/comments/wrxcz0/probably_an_unpopular_opinion_but_i_think_a_lot/)
- <https://corq.studio/insights/the-cult-of-glossier-why-the-brands-tuned-in-community-is-turning-on-the-company-over-alleged-product-changes/>
- <https://explodingtopics.com/blog/beauty-trends>

Thankyou!

